

DATA 3464: Fundamentals of Data Processing

Missing and Weird Data

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Topic overview

- Written test 1 info
- What to do with missing data
- Detecting and handling outliers

Resources used:

- [Feature Engineering Chapter 8](#)
- Hands on Machine Learning with Scikit-Learn and Tensorflow/PyTorch, Chapter 2. Available at [MRU Library](#)
- [Scikit-learn user guide: Chapters 2 and 7](#)

Written test 1 info

- 1 **handwritten** reference page on 8.5x11" paper, single-sided
- A non-programmable emotional support calculator
- 6 questions MC, 9 TF, 3x long answer (similar to class practice)
- Total of 50 points (1 point per MC), 80 minutes
- Last semester's midterm is on D2L — it has concepts we haven't covered yet!
- Two tests worth 35% total — this one is 15% or 20%, whichever is best for you

Written test 1 content

What is *not* on the test:

- Clustering, because we skipped it
- Detailed knowledge of gradient descent or probability distributions
- Any kind of Python syntax or coding

What *is* on the test?

- Concepts from lectures, labs, and assignments
- Up to and including basic imputation of missing values (today!)
- More emphasis on things you did in labs/assignments

The problem

- As you've seen, real-world data is messy
- Missing values are common, other values don't make sense
- We need to decide how to deal with these problems

What examples have we seen so far?

Why might data be missing or weird?*

Missing data

When data are missing in the *features* we have a few options:

0. Do nothing! Some algorithms (e.g. [Decisions Trees](#)) can handle missing values
1. Remove *features* with missing values
2. Remove *samples* with missing values
3. Invent a new value to represent "missingness"
4. **Impute** a value based on other data

Most important: understand why data are missing (more EDA!)

Option 1: removing features

- If a feature has:
 - A high proportion of missing values, *and*
 - Little apparent relationship to the target *or*
 - its information is redundant with other features
- It may be reasonable to remove it entirely. You can drop it, e.g.:

```
df.drop(columns=['feature_name'], inplace=True)
```

or (probably more reliable) just not select it when building your pipeline

Option 2: removing samples

- If:
 - Only a small number of samples have missing values
 - The values are missing completely at random

Then you can drop them **from the training data**:

```
train_nona = train.dropna(subset=[["features", "we", "care", "about"]])
```

- Good idea if the same samples have missing values from multiple features
- Still useful to explore *why* data are missing

What does this imply at inference time?

Option 3: invent a new value

- Categorical features: add a new category for "missing"
- Add a new binary feature indicating whether the value was missing
- I have seen advice to use extreme values for numerical features, like the -1 income in the OKCupid dataset, but I'm not convinced this is a good idea

Case study: where missingness is informative

Option 4: impute missing values

- Fill in the missing values with an "educated guess"
- Replace missing value with:
 - constant
 - mean, median, or mode (`most_frequent`)
- Use other features to infer missing value:
 - K-nearest neighbours
 - simple models to predict missing values
- Can be combined with option 3 to [indicate missing features](#)
- How much to impute? [Feat.Engineering](#) suggests no more than 20%

Choosing an imputation strategy

Strategy	When to use
Constant	When there is a reasonable default value
Mean	Numeric features with normal distribution
Median	Numeric features with extreme outliers
KNN	Relationship with other features
Missingness indicator	If missingness seems informative

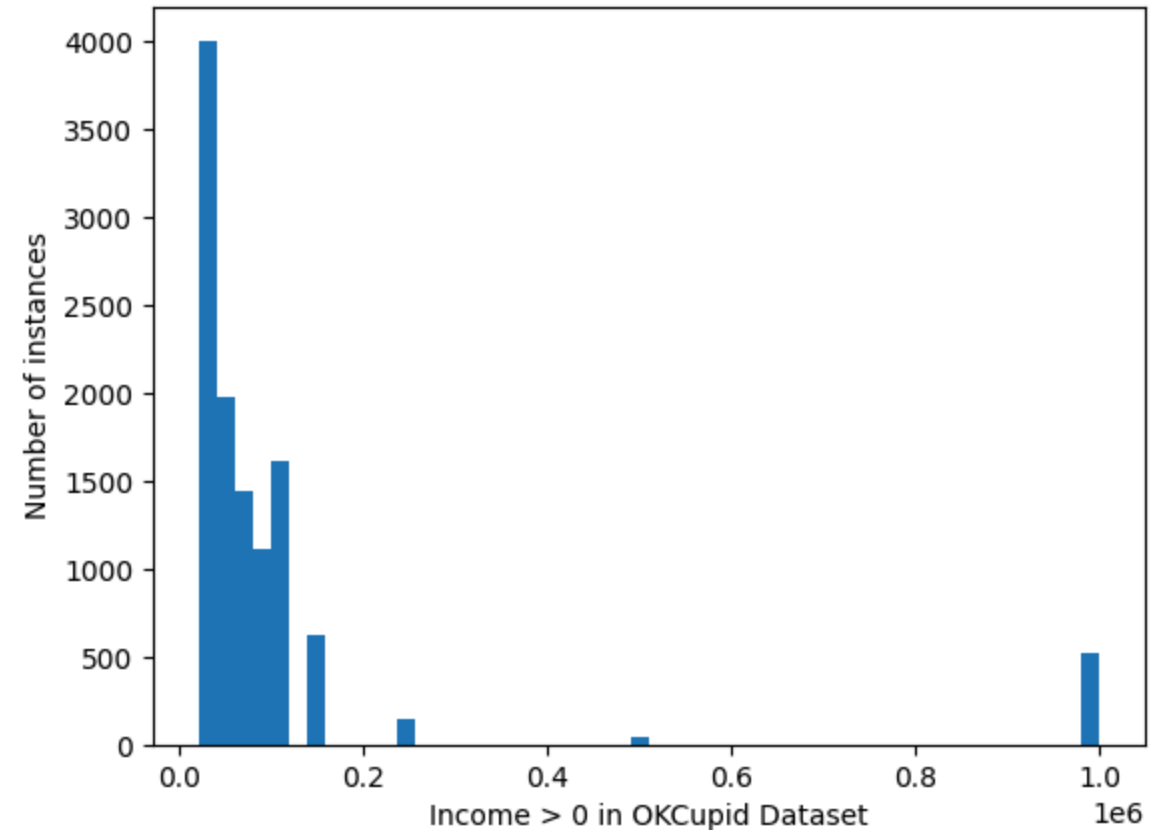
Outliers

An outlier is an observation which deviates so much from the other observations as to arouse suspicions that it was generated by a different mechanism. — D. M. Hawkins

- There are many entire books dedicated to outlier detection
- Useful for anomaly detection, e.g:
 - fraud detection
 - failure prediction
 - **Many other applications**
- Our focus is on dealing with outliers in preprocessing

Detecting outliers

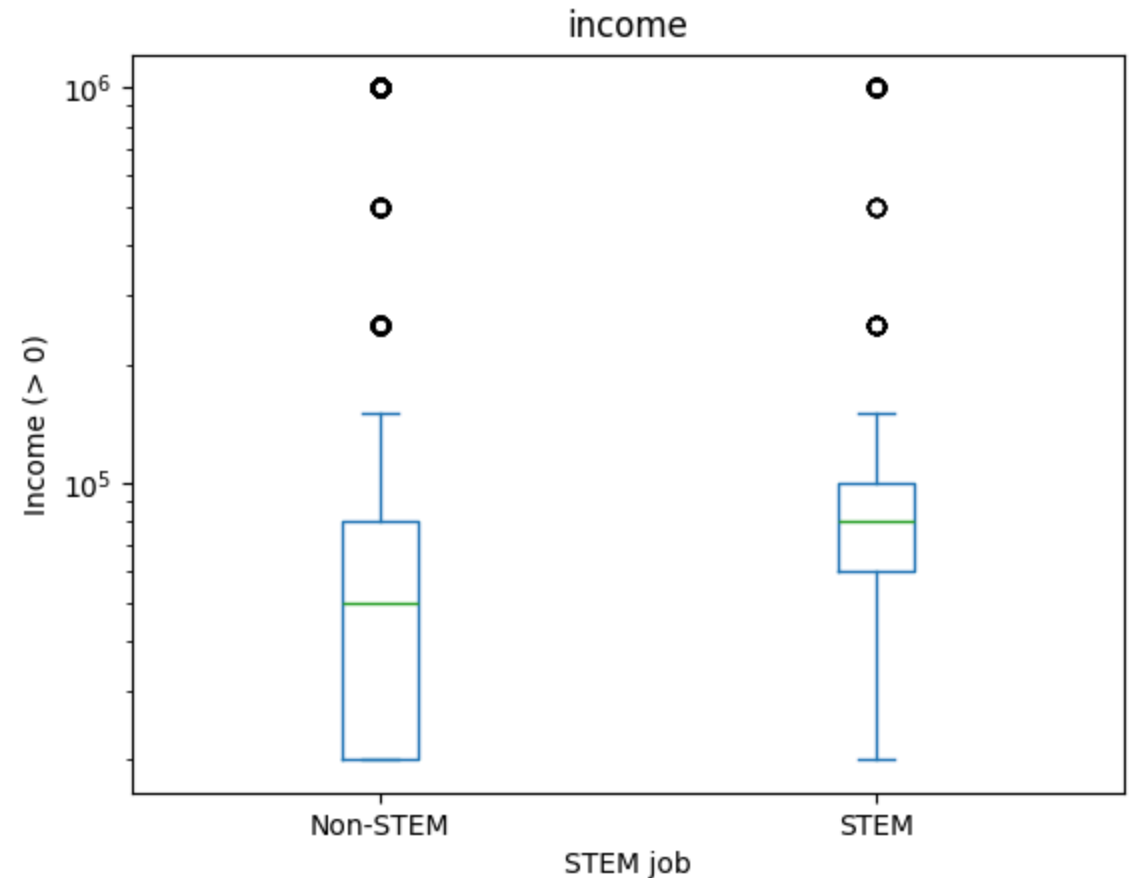
- Visually as part of EDA
- Statistically, e.g. $> 3\sigma$
- Algorithmically, e.g. [Isolation Forests](#)
- As usual, context and domain knowledge are essential



Context matters!

- Look at the relationship between outliers and target (training dataset, of course)

What do the dots on box plots represent?



What to do about outliers?

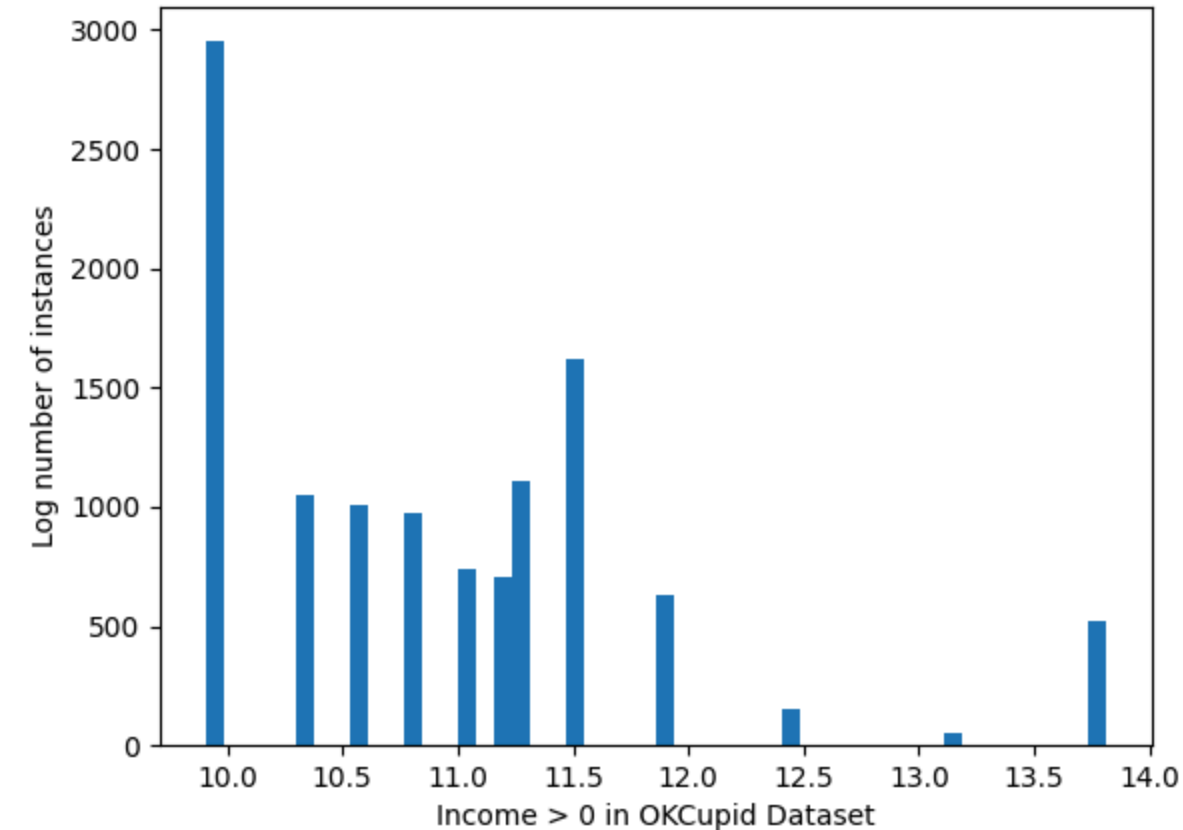
As usual, very data- and model-dependent. Tree based methods are particularly impacted by outliers!

1. Data transformations
2. Drop the samples
3. Encode them somehow
4. Leave them alone

Any other ideas?

Nonlinear transformations

- Transforming the data does not actually remove the outlier
- Can help make the relationship less extreme



Dropping the samples

- In general, not a thing I love to do
- If you drop a sample from training, you need to decide what to do at inference
- My opinion: Only do it if you're confident it's an error in the dataset

Can you think of an example?

- What can you do at inference time when outliers are encountered?

Encoding outliers

A few other options that might fall into "encoding":

- Just like with missing values, binary column indicating outlier/inlier
- Remove the values and convert them to missing
- Bin or impose a cap (floor/ceiling) on the value
- Replace the numeric value with a rank or quantile bucket
- Probably other things!

Leave them alone

If your outliers are:

- Real values (not data entry or other errors)
- Representative of things that might happen during inference

Then you probably want to keep them!

Consider using a [RobustScalar](#) to standardize if you have lots of outliers

Missing values and outliers in the target

- If you have missing values in the target, you probably want to avoid imputing
- This is a good case for dropping samples from **training**!
- Outliers are trickier — again, check if they're real or mistakes
- There may be a case for transforming the target

Coming up next

- Interactions between variables
- Feature selection
- Written exam 1, assignment 2
- Reading week!!!